

Detailed modulation response analyses on enhanced single-mode QWS-DFB lasers with distributed coupling coefficient

Xin-Hong Jia ^{a,*}, Dong-Zhou Zhong ^b, Fei Wang ^c, Hai-Tao Chen ^d

^a School of Physics and Electronic Engineering, Sichuan Normal University, Chengdu 610068, People's Republic of China

^b School of Informatics, Wuyi University, Jiangmen 529020, China

^c School of Mathematica Scientia, Chongqing Institute of Technology, Chongqing 400050, China

^d Fundament Department, The Chinese People's Armed Police Force Academy, Langfang 065000, China

Received 6 April 2007; received in revised form 2 May 2007; accepted 6 May 2007

Abstract

By using the coupled mode equations, the dynamic characteristics of $\lambda/4$ phase-shifted distributed feedback laser diode with distributed coupling coefficient (QWS-DCC-DFB) is analyzed. The numerical simulation indicates that, compared with purely QWS-DFB laser, the better dynamic single-mode can be achieved by QWS-DCC-DFB laser; Under the smaller biasing current, the modulation band-width is narrower, this difference shrinks for larger biasing current; For large-signal modulation, the distributed coupling coefficient is helpful to increase the output extinction ratio, but worsens the wavelength chirping. The good dynamic characteristic can be obtained by selecting system parameters reasonably. QWS-DCC-DFB laser would have potentials in optics communications because of better dynamic SMSR.

© 2007 Elsevier B.V. All rights reserved.

Keywords: $\lambda/4$ phase-shift; Distributed feedback laser diode; Distributed coupling coefficient; Dynamic single-mode suppression ratio; Modulation response; Large-signal analysis

1. Introduction

Semiconductor laser diode (SLD) has been widely used in wavelength-division-multiplexing (WDM) network due to small size, low power consumption, fast response, and so on [1]. In high-speed optical communication system, the most important characteristics required for SLD is the stable dynamic single-mode operation (DSM) [2,3]. The $\lambda/4$ phase-shifted distributed feedback (QWS-DFB) laser diode is the good candidate to achieve stable single-mode operation because of high side-mode suppression ratio (SMSR) (often SMSR higher than 30 dB can be realized) [3–6]. However, this better SMSR only exists near threshold, and under larger biasing current, the severe spatial hole-burning effect gives rise to SMSR degradation

[3,4]. Usually two methods are proposed to solve this issue: The first is to weaken the nonuniform distribution of carriers, which can be obtained by introducing longitudinal chirped grating for effective index or Bragg period (GC-DFB), or by utilizing multiple discrete phase-shift (MPS-DFB) [7,8]. The second is to raise the gain margin by introducing gain coupling mechanism [1,2,9–11] or constructing distributed coupling coefficient grating (DCC-DFB).

For DCC-DFB lasers, the large SMSR can be kept even for very-high biasing current compared with other nonuniform DFB lasers [12]. However, earlier studies on QWS-DCC-DFB lasers mostly were carried out under quasi-continuous-wave (CW) hypothesis. It is essential to further explore its small-signal modulation response since it is criterion to evaluate dynamic performance of SLD. On the other hand, for the case of larger-signal modulation, the output light accompanies with wavelength chirp owing to the refraction index fluctuation inside active-layer. The

* Corresponding author.

E-mail address: jxh_0@yahoo.com.cn (X.-H. Jia).

optical signal with wavelength chirp would suffer from dispersion penalty during its transmission in fiber. The large-signal modulation can also bring about pulse distortion and extinction ratio (EO) decrease, eventually impact the bit-error-ratio (BER) performance. Based on the above consideration, by using the coupled mode equations, this paper presents a detailed theoretical investigation on the influences of system parameters on small-signal modulation response, and output waveform, wavelength chirp under large-signal modulation. The related results may provide an instructive insight and are beneficial for device design and optimization.

2. Theoretical model

The schematic diagram of QWS-DCC-DFB lasers is shown in Fig. 1. The $\lambda/4$ phase-shift is located at middle of waveguide layer, and the variable coupling coefficient takes step-like distribution along longitudinal direction.

2.1. Time-domain coupled mode equations

The optical field E in active-layer for DFB lasers can be decomposed into two terms

$$E(z, t) = [A_f(z, t) \exp(i\beta_0 z) + A_b(z, t) \exp(-i\beta_0 z)] \times \exp(-i\omega_0 t) + c.c. \quad (1)$$

where A_f , A_b are the slowly varying amplitudes of forward and backward waves, $\beta_0 = \pi/\Lambda$ is Bragg wave-number, Λ is grating period, ω_0 is the referenced central frequency.

Substituting (1) into wave equation, and under slowly variable amplitude approximation, the evolution of forward and backward waves obeys the following time-domain coupled mode equations [13]:

$$\frac{\partial A_f}{\partial z} + \frac{1}{v_g} \frac{\partial A_f}{\partial t} = i[\Delta\beta A_f + k A_b] + S_f \quad (2a)$$

$$\frac{\partial A_b}{\partial z} - \frac{1}{v_g} \frac{\partial A_b}{\partial t} = -i[\Delta\beta A_b + k A_f] + S_b \quad (2b)$$

where v_g is the light group velocity in the medium, $\Delta\beta$ is the complex detuning, defined by

$$\Delta\beta = \delta - i\frac{\Gamma g}{2}(1 - i\alpha) + i\frac{\alpha_{\text{int}}}{2} \quad (3a)$$

with

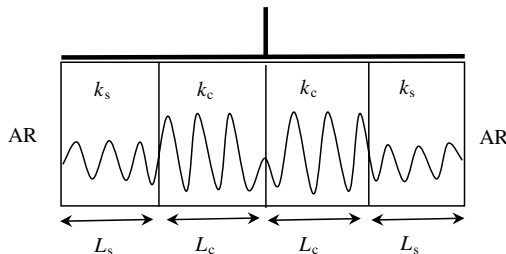


Fig. 1. Schematic diagram of QWS-DCC-DFB lasers.

$$\delta = n_0 \frac{\omega_0}{c} - \beta_0 \quad (3b)$$

where δ is the initial detuning, Γ is the confinement factor, c is the light velocity in vacuum, n_0 is the effective mode refractive index without pumping, α is the line-width enhancement factor, g and α_{int} account for the power gain and internal loss coefficients, respectively. The effect of spontaneous emission is represented by two statistically independent white-Gaussian distributed random process S_f , S_b , which autocorrelation function is written as [13]

$$\langle S_{f(b)}(z, t) S_{f(b)}^*(z', t') \rangle = EA_{\text{cross}}(\Gamma\beta BN^2)/L\delta(z - z')\delta(t - t') \quad (4)$$

where A_{cross} is the cross-sectional area of active-layer, E is the photon energy, B is the bimolecular radiative recombination coefficient, N is the carrier density, L is the length of active section, and β is the spontaneous coupling factor.

For our discussed QWS-DCC-DFB lasers, the coupling coefficient k variation takes the form of

$$kL = 2(k_c L_c + k_s L_s) \quad (5a)$$

$$\frac{k_c L_c}{k_s L_s} = r \quad (5b)$$

where $2k_c L_c$ and $2k_s L_s$, respectively, characterize the normalized coupling coefficient for central and side sections, r is their ratio. For simplicity, each section length is assumed to equal to $L/4$.

To avoid the dual-mode degeneration, the coated anti-reflection (AR) layer at the end fact of DFB lasers is assumed [14]. Under this circumstance, the boundary conditions of forward and backward waves can be described by

$$z = 0 : A_f(0, t) = 0, \quad A_r(0, t) = A_b(0, t) \quad (6a)$$

$$z = L : A_b(L, t) = 0, \quad A_t(L, t) = A_f(L, t) \quad (6b)$$

where A_r and A_t are the slowly variable amplitude for backward and forward output waves, respectively.

2.2. Carrier rate equation and gain model

The carrier rate equation can be described by [14]

$$\frac{\partial N}{\partial t} = \frac{I}{qV} - \frac{N}{\tau_s(N)} - \frac{\Gamma g(|A_f|^2 + |A_b|^2)}{A_{\text{cross}} E} \quad (7a)$$

where I is injection current, q is electron charge, V is active-layer volume, and the carrier lifetime $\tau_s(N)$ is defined as [14]

$$\tau_s(N) = (A + BN + CN^2)^{-1} \quad (7b)$$

where A , C are the non-radiative recombination and Auger recombination coefficients. The dependence of material gain coefficient g on carrier density and wavelength can be approximated to [15]

$$g(N, \omega) = \frac{a_0(N - N_0)}{1 + \varepsilon S} \left\{ 1 - \left[\frac{\omega - \omega_{\text{peak}}(N)}{\Delta\omega_g(N)} \right]^2 \right\} \quad (8a)$$

where a_0 is the gain constants, N_0 is carrier density at transparency, $S = (|A_f|^2 + |A_b|^2)/(v_g A_{\text{cross}} E)$ is the intra-cavity photon density, ε is used to represent the nonlinear gain compression mechanism including carrier density pulsation (CDP), carrier heating (CH), and spectral hole-burning (SHB). ω_{peak} and $\Delta\omega_g$ are the gain peak frequency and gain band-width, which is assumed to shift linearly with carrier density according to

$$\omega_{\text{peak}}(N) = \omega_{\text{tr}} + C_{\omega}(N - N_0) \quad (8b)$$

$$\Delta\omega_g(N) = C_{\omega}(N - N_0) \quad (8c)$$

where ω_{tr} is the gain peak frequency at transparency, C_{ω} denotes the variation slope of gain peak frequency and gain band-width with respect to carrier density.

2.3. Numerical simulation algorithm

The transfer-matrix method (TMM) [16,17] is utilized to simulate dynamic behavior of wave-length conversion in DFB lasers. Split the length L into M equal sections from input end, assume each section is uniform, and label the localized amplitudes before and after each section as $A_{f,i+1}(A_{b,i+1})$ and $A_{f,i}(A_{b,i})$ ($i = 1, 2, \dots, M$), respectively, then

$$\begin{bmatrix} A_{f,i+1} \\ A_{b,i+1} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} A_{f,i} \\ A_{b,i} \end{bmatrix} \quad (9a)$$

where the detailed transferred matrix formula is similar to Ref. [18] given by the authors. For the phase-shifted DFB lasers, the transfer-matrix at phase-shifted location should be multiplied by the matrix F_p [18]

$$T_p = \begin{bmatrix} \exp(i\psi_{\text{sh}}) & 0 \\ 0 & \exp(-i\psi_{\text{sh}}) \end{bmatrix} \quad (9b)$$

where ψ_{sh} represents the phase-shifted magnitude. If time step size Δt is related to spatial step size Δz as $\Delta t = \Delta z/v_g$, the resulting update format takes the form of

$$\begin{bmatrix} A_{f,i+1}(t + \Delta t) \\ A_{b,i}(t + \Delta t) \end{bmatrix} = \begin{bmatrix} a_{11}(t) - \frac{a_{12}(t)a_{21}(t)}{a_{22}(t)} & \frac{a_{12}(t)}{a_{22}(t)} \\ -\frac{a_{21}(t)}{a_{22}(t)} & \frac{1}{a_{22}(t)} \end{bmatrix} \begin{bmatrix} A_{f,i}(t) \\ A_{b,i+1}(t) \end{bmatrix} \quad (9c)$$

To take the frequency-dependent gain response into account, we treat the gain term in (9) by using finite impulse response digital filter (FIR-DF) [14]. Starting from the initial condition, using (2)–(6) and (9), the longitudinal distributions of optical fields can be computed. The distribution of gain coefficient at next time can be calculated from (7) and (8) by means of fourth-order Runge–Kutta method. Repeating these steps, the whole output waveforms can be obtained. It should be stressed that the precision and validity can be improved by increasing M . Our simulation results have shown that for conventional active section length ($\sim 300 \mu\text{m}$) and data rate (less than $\sim 40 \text{ Gb/s}$), if M closes to 100 or more, the output optical and carrier distributions remain unchanged.

3. Results and discussion

The device structure and material parameters used in simulation are listed in Table 1. In the following discussions, the performance comparison between QWS-DCC-DFB and QWS-DFB lasers under various system parameters will be reviewed in detail.

3.1. Dynamic single mode characteristics

Fig. 2 shows the normalized threshold gain coefficient $g_{\text{th}}L$ versus output mode (the calculation method can be found in Ref. [19]), where $kL = 2$, the unfilled square and filled dot, respectively, correspond to QWS-DFB and DCC-QWS-DFB lasers. It can be seen that, although the lasing threshold gain of QWS-DCC-DFB lasers (~ 6.6) is slightly higher than QWS-DFB lasers (~ 5.1), but the first side-mode threshold gain for QWS-DCC-DFB lasers (~ 16.1) is far bigger than QWS-DFB lasers (~ 9.4). This implies that the side-mode energy is weakened efficiently, and the single-mode range is broadened significantly after using distributed coupling coefficient structure.

Fig. 3a and b illustrate the transient response and output optical spectrum for QWS-DFB lasers, where $kL = 2$, the biasing current takes the form of step-like function, and its final state value is $I = 90 \text{ mA}$. The optical spectrum is obtained by performing fast Fourier transform (FFT) for output optical field within $4 \sim 5 \text{ ns}$. To facilitate comparison, the transient response and output optical spectrum

Table 1
Device structure and material parameters used in calculations

Symbol	Parameter	Value
L	Active-layer length	$300 \mu\text{m}$
A_{cross}	Active-layer cross-sectional area	$2 \times 10^{-13} \text{ m}^2$
V	Active-layer volume	$6 \times 10^{-17} \text{ m}^3$
Γ	Confinement factor	0.35
λ_B	Bragg wavelength	1550 nm
λ_0	Referenced central wavelength	1550 nm
ψ_{sh}	Phase-shifted magnitude	$\pi/2$
n_0	Refractive index of active-layer	3.3
v_g	Group velocity	$7.5 \times 10^7 \text{ m/s}$
N_0	Carrier density at transparency	
α_{int}	Internal power loss	$1.1 \times 10^{24} \text{ m}^{-3}$
a_0	Material gain constant	$4 \times 10^3 \text{ m}^{-1}$
A	Non-radiative recombination coefficient	$2.5 \times 10^{-20} \text{ m}^2$
B	Bimolecular radiative recombination coefficient	$1.5 \times 10^8 \text{ s}^{-1}$
C	Auger recombination coefficient	$2.5 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$
α	Line-width enhancement factor	$9.4 \times 10^{-41} \text{ m}^6 \text{ s}^{-1}$
β	Spontaneous coupling factor	4.86
r	Ratio of normalized coupling coefficient for central and side sections	1×10^{-5}
$\omega_{\text{tr}}/(2\pi)$	Gain peak frequency at transparency	3
$C_{\omega}/(2\pi)$	Variation slope of gain peak frequency and gain band-width with respect to carrier density	$1.87 \times 10^{14} \text{ Hz}$
ε	Gain compression factor	$6.4 \times 10^{-12} \text{ m}^3/\text{s}$ $2 \times 10^{-23} \text{ m}^3$

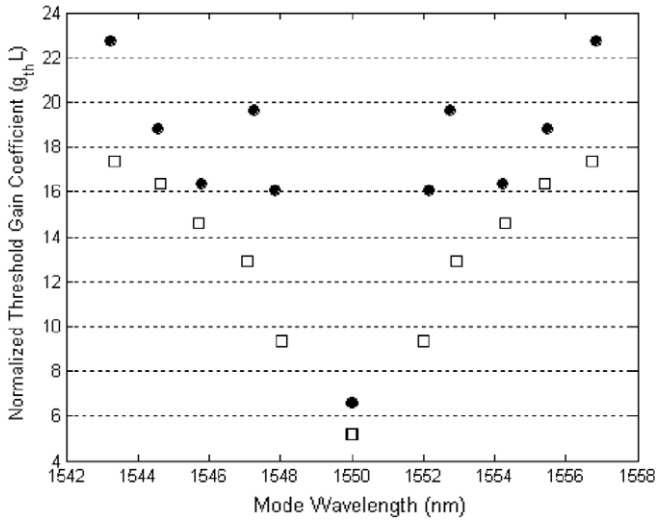


Fig. 2. Normalized threshold gain coefficient $g_{th}L$ versus output mode, where the unfilled square and filled dot, respectively, correspond to QWS-DFB and DCC-QWS-DFB lasers, $kL = 2$.

of QWS-DCC-DFB lasers are also displayed in Fig. 4a and b. Our simulation shows that, if the biasing current is lower than ~ 90 mA, both of QWS-DCC-DFB and QWS-DFB lasers show better dynamic single-mode characteristics. But when the biasing closes to ~ 90 mA, the lasing output of QWS-DFB lasers experiences such process as transient oscillation, stable output and finally to oscillation state again. The oscillation output stems from beating frequency of multiple mode lasing. From the inset in Fig. 3a, it can be observed that, this oscillation frequency is ~ 200 GHz, which corresponds to the central frequency difference between main-mode and the first side-mode. The SMSR is estimated to be ~ 21 dB from Fig. 3b. If the biasing current is further increased, the SMSR would deteriorate rapidly. However, for QWS-DCC-DFB lasers, the side-mode is almost suppressed completely, and good dynamic single-mode is maintained, which can be confirmed from Fig. 4b. For $kL = 2$, our calculation demonstrates that SMSR larger than 30 dB can be maintained for QWS-DCC-DFB lasers when biasing current is within ~ 200 mA.

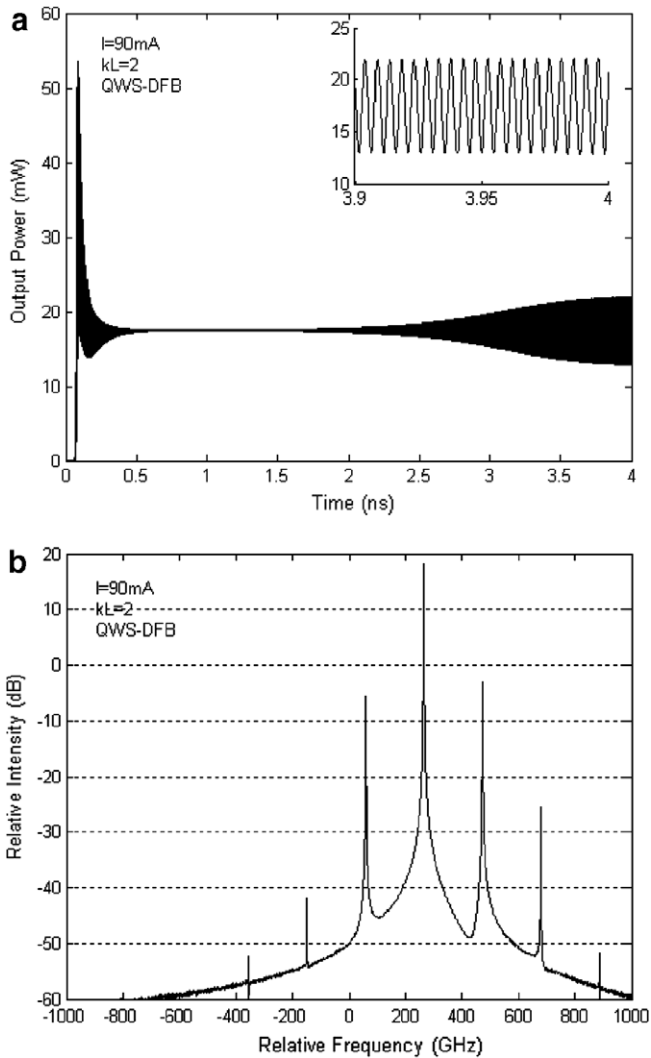


Fig. 3. The (a) transient response and (b) output optical spectrum for QWS-DFB lasers, where $kL = 2$.

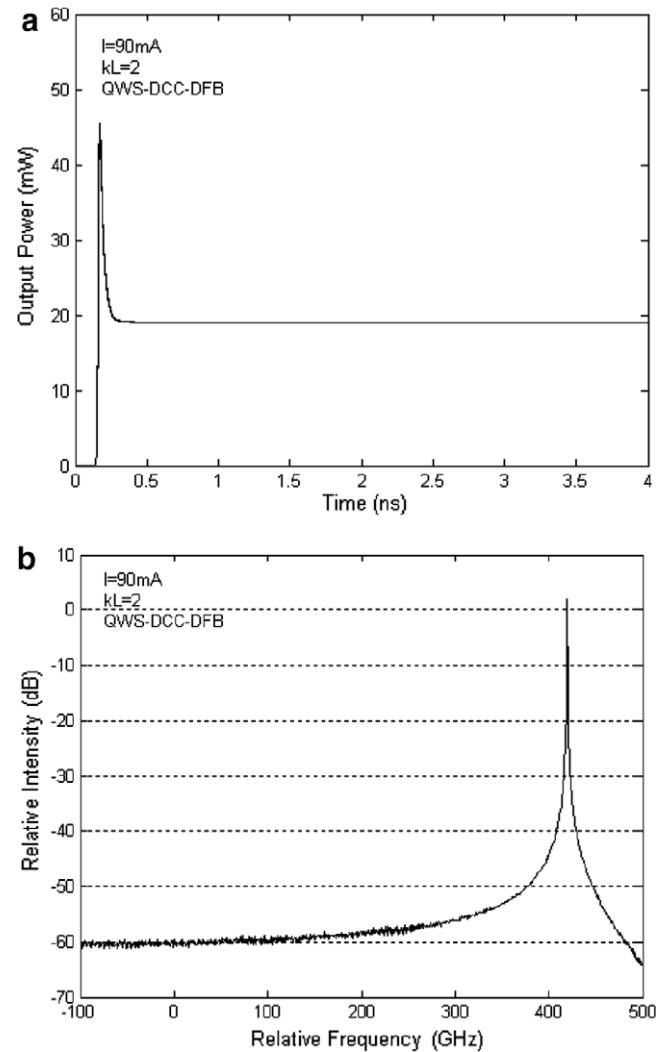


Fig. 4. The (a) transient response and (b) output optical spectrum for QWS-DCC-DFB lasers, where $kL = 2$.

3.2. Small-signal modulation response

Fig. 5a and b, respectively, show the small-signal modulation response for QWS-DFB and QWS-DCC-DFB lasers under various biasing currents, where $kL = 2$, the biasing current I_d is selected to ensure single-mode operation. With the increased biasing current, the 3 dB modulation band-width is broadened, which is caused by the increased relaxation oscillation. For smaller biasing current (~ 40 mA), the 3 dB modulation band-width of QWS-DCC-DFB lasers is ~ 8 GHz, which is lower than QWS-DFB lasers (~ 9 GHz). The modulation band-width difference is more visible if the biasing current is further decreased. This phenomenon can be explained by noting that, for QWS-DCC-DFB lasers, the suppressed side-mode makes the intra-cavity energy smaller, and the laser shows obvious LSHB effect resulted from distributed coupling coefficient structure (see next discussion), so the relaxation oscillation is weakened.

In addition, the 3 dB modulation band-width of QWS-DFB lasers is kept to be ~ 11 GHz for 40–60 mA biasing current, whereas the modulation band-width of QWS-DCC-DFB lasers is kept to be ~ 11 –12 GHz. This means that under larger biasing current, both of their modulation band-widths tend to saturate, which makes the difference shrink with the increased biasing current. The answer is that, with the increased biasing current, the raised gain increases intra-cavity photon density, which tends to intensify relaxation oscillation. Meantime, the increased photon density can weaken relaxation oscillation via gain compression mechanism in turn. It is this balance function that saturates modulation band-width.

Altering the coupling coefficient, Fig. 6a and b, respectively, show the small-signal modulation response for QWS-DFB and QWS-DCC-DFB lasers under various biasing currents, where $kL = 3$. Both of their modulation response are similar to the case of $kL = 2$ except that single-mode range is narrowed. This can attributed to the fact

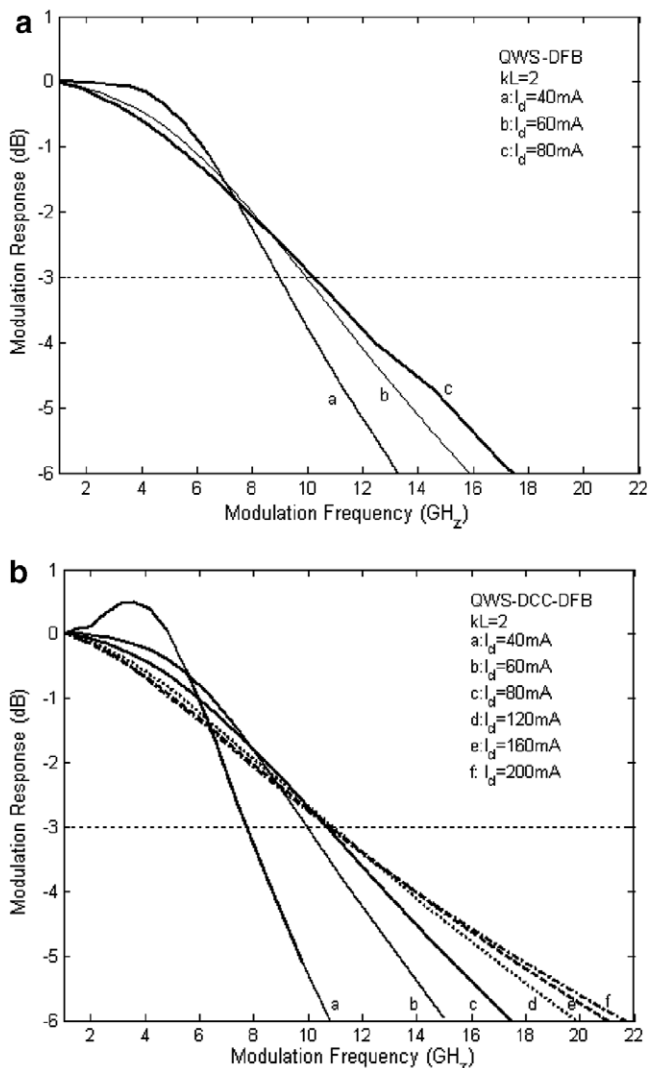


Fig. 5. The small-signal modulation response for (a) QWS-DFB and (b) QWS-DCC-DFB lasers under various biasing currents, where $kL = 2$.

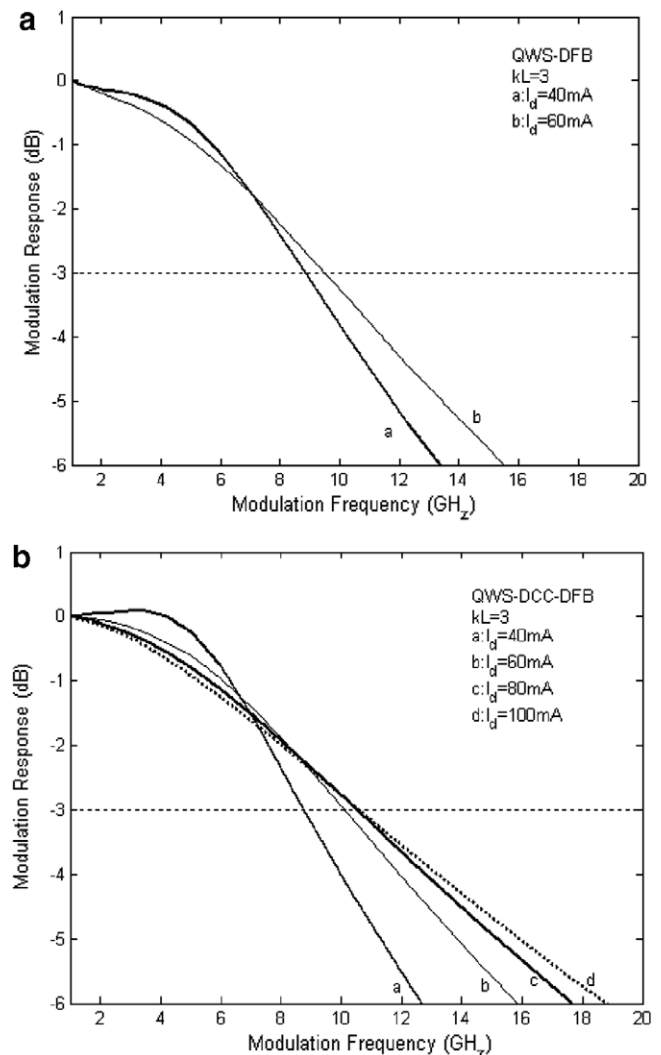


Fig. 6. The small-signal modulation response for (a) QWS-DFB and (b) QWS-DCC-DFB lasers under various biasing currents, where $kL = 3$.

that larger coupling coefficient strengthens the interaction not only between forward and backward waves, but also between adjacent modes, therefore, the multiple mode operation can be reached easily.

3.3. Large-signal modulation response

Fig. 7a and b, respectively, depict the injection current and output power waveforms for QWS-DFB and QWS-DCC-DFB lasers under large-signal modulation, where $kL = 2$, the biasing current $I_b = 35$ mA, peak modulation current $I_p = 20$ mA, and bit rate $R = 5$ Gb/s. In calculation a $2^7 - 1$ pseudorandom bit sequence (PRBS) with nonreturn-to-zero (NRZ) pattern is chosen as input data stream. To avoid space bit (“0”) is under spontaneous emission state, the biasing current should approach to around lasing threshold at least. It can be seen that, for space bit (“0”), the power output of QWS-DCC-DFB lasers is lower than QWS-DCC lasers, but for mark bit (“1”), their power outputs almost have no difference, which means that the

extinction ratio of QWS-DCC-DFB lasers is better than purely QWS-DFB lasers. To explore the concealed physical origin, Fig. 8a and b, respectively, show the intra-cavity total power and carrier density steady distribution for QWS-DFB and QWS-DCC-DFB lasers under various biasing currents, where $kL = 2$. For QWS-DFB lasers, the intra-cavity energy tends to accumulate to central section with increased biasing current, and consequently the carrier density tends to build up to side section (see curves a1 and a2). But for QWS-DCC-DFB lasers, the intra-cavity energy tends to accumulate to side section with increased biasing current, moreover, the considerable discontinuous behavior happens near break point of coupling coefficient, and under this case, the carrier density tends to build up to central section (see curves b1 and b2). As a result, for larger current injection (i.e., mark bit), the power outputs of both lasers have no evident difference. In addition, for lower current injection (space bit), the power out-

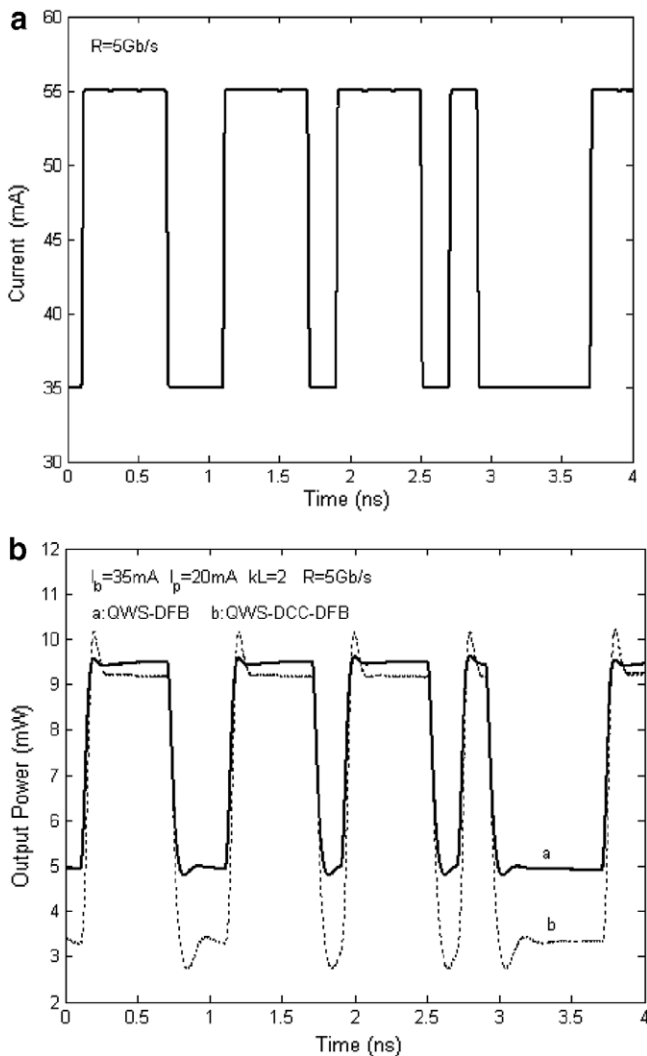


Fig. 7. The (a) injection current and (b) output waveform for QWS-DFB and QWS-DCC-DFB lasers under large-signal modulation, where $kL = 2$.

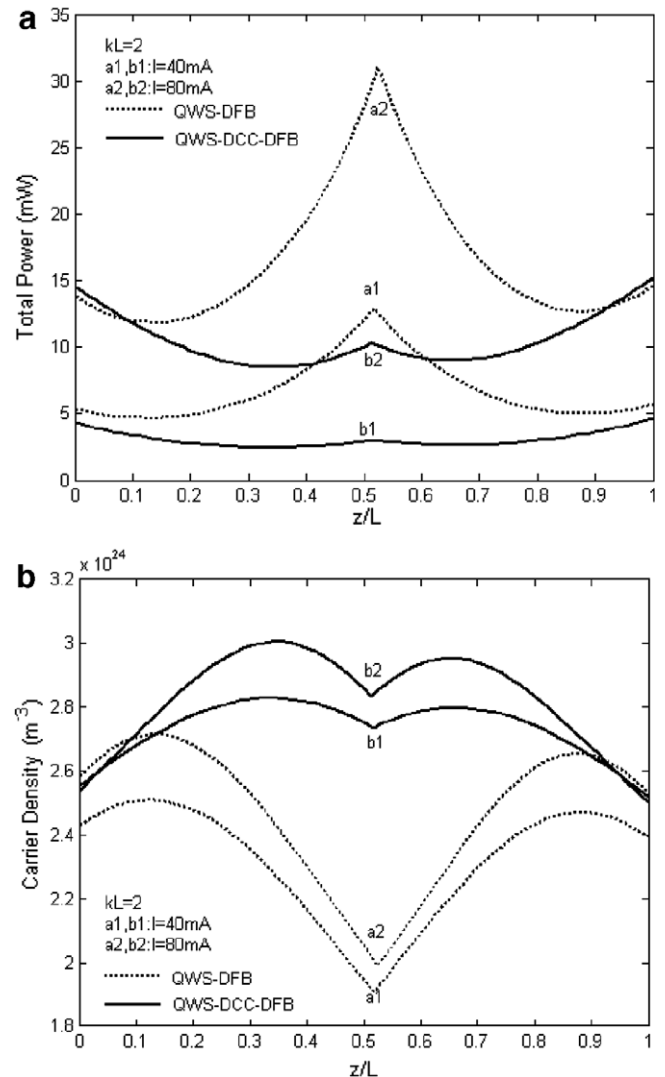


Fig. 8. The (a) intra-cavity total power and (b) carrier density steady distribution for QWS-DFB and QWS-DCC-DFB lasers under various biasing currents, where $kL = 2$.

put of QWS-DCC-DFB lasers is lower than QWS-DCC lasers owing to suppressed side-mode energy.

Fig. 9a shows the variation of lasing wavelength with time for QWS-DFB and QWS-DCC-DFB lasers, where the used data is the same as Fig. 7. At the rising-edge of mark (“1”), the negative wavelength chirp is observed, which comes from refraction index decrease of active-layer because of supplied carrier density. Opposite result appears for falling-edge. Additionally, under the parameters given above, the peak-peak wavelength of QWS-DCC-DFB lasers is ~ 0.17 nm, while only ~ 0.06 nm is found for QWS-DFB lasers.

Fig. 9b plots the variation of peak-to-peak wavelength chirp with peak modulation current for QWS-DFB and QWS-DCC-DFB lasers, where $kL = 2$, $I_b = 35$ mA, $R = 5$ Gb/s. It can be seen that, the wavelength chirp of both lasers worsens with increased peak modulation current and exhibits saturation trend. The worsened wavelength chirp can be explained that the increased biasing current leads to evident refraction index fluctuation. But if the biasing current is raised to some degree, gain saturation and nonlinear gain compression make the fluctuation of refraction index and lasing wavelength be stabilized.

From this diagram, it can also be seen that, the wavelength chirp of QWS-DCC-DFB lasers is always higher than QWS-DFB lasers, which is more and more apparent with increased peak modulation current. As mentioned above, the carrier density of QWS-DCC-DFB lasers shows discontinuous behavior near the break point of coupling coefficient. This LSHB effect can bring about the refraction index fluctuation in spatial domain. The spatial fluctuation of refractive index would inevitably lead to its change in time-domain, and finally result in larger wavelength chirp.

Fig. 9c shows the variation of peak-to-peak wavelength chirp with bit rate for QWS-DFB and QWS-DCC-DFB lasers, where $kL = 2$, $I_b = 35$ mA, $I_p = 20$ mA. It can be seen that, if the bit rate is limited to be smaller than ~ 10 Gb/s, the chirp ripple is slight. Beyond this range, the wavelength chirp decreases, which is caused by finite carrier lifetime (typically 0.2–2 ns).

Fig. 10 gives the line-width as a function of biasing current for QWS-DFB and QWS-DCC-DFB lasers under con-

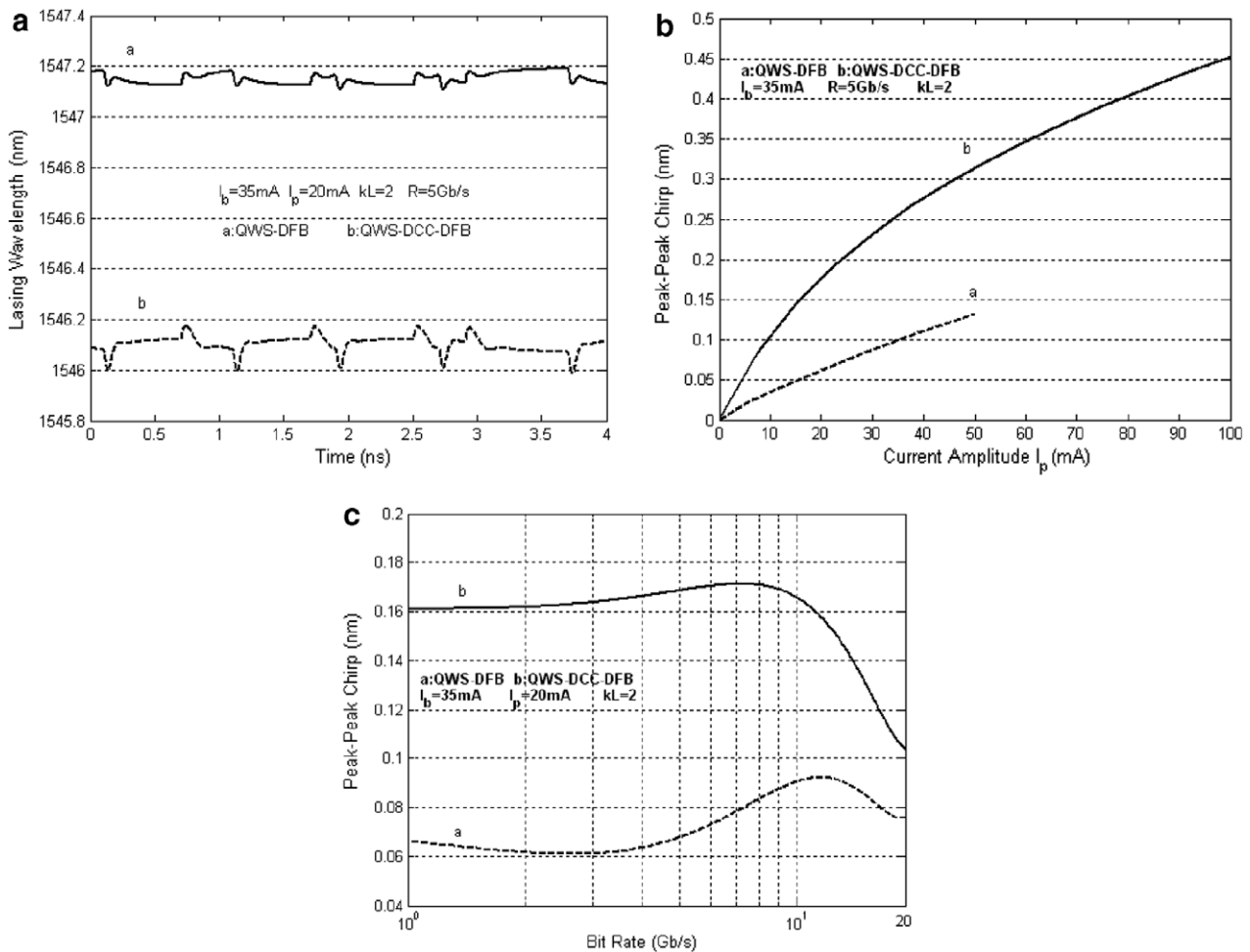


Fig. 9. (a) Variation of lasing wavelength with time for QWS-DFB and QWS-DCC-DFB lasers, where $kL = 2$. (b) Variation of peak-to-peak wavelength chirp with peak modulation current for QWS-DFB and QWS-DCC-DFB lasers, where $kL = 2$. (c) Variation of peak-to-peak wavelength chirp with bit rate for QWS-DFB and QWS-DCC-DFB lasers, where $kL = 2$.

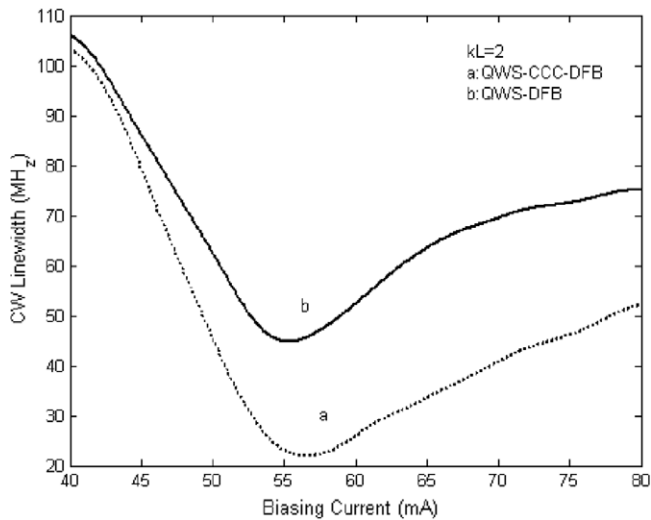


Fig. 10. line-width as a function of biasing current for QWS-DFB and QWS-DCC-DFB lasers under continuous-wave (CW) condition, where $kL = 2$.

tinuous-wave (CW) condition, where $kL = 2$. It can be seen that, if biasing current is smaller than a critical value, both of their line-widths decrease with increased biasing current because of suppressed spontaneous emission by simulated emission. But under large biasing current, the line-width is increased gradually owing to nonlinear gain [14]. Additionally, QWS-DCC-DFB laser has narrower line-width, especially for higher biasing current injection.

4. Conclusions

In summary, the modulation characteristic analysis of QWS-DCC-DFB lasers has been carried out by using coupled mode equations. The numerical simulation shows that, compared with purely QWS-DFB laser, the better dynamic single-mode can be achieved by QWS-DCC-DFB laser. Under the smaller biasing current, the modulation band-width is narrower, this difference shrinks for larger biasing current; For large-signal modulation, the distributed coupling coefficient is helpful to increase the output extinction ratio, but worsens the wavelength chirping. The modulation performance of QWS-DCC-DFB

lasers can be improved by: (1) To make the wavelength chirp, extinction ratio, and modulation band-width optimized under the precondition of larger dynamic SMSR, the biasing current and coupling coefficient must be selected reasonably; (2) To reduce wavelength chirp, the medium with smaller line-width enhancement factor should be utilized as active-layer material, this can be realized by introducing multiple quantum well (MQW), etc., [14].

Acknowledgement

The authors acknowledge greatly the reviewers for their helpful comments.

References

- [1] D.A. Cardimona, M.P. Sharma, V. Kovanis, A. Gavrielides, *IEEE J. Quantum Electron.* 31 (1) (1995) 60.
- [2] J.Y. Chen, A. Champagne, R. Maciejko, T. Makino, *IEEE J. Quantum Electron.* 33 (1) (1997) 33.
- [3] P. Correc, *IEEE J. Quantum Electron.* 30 (11) (1994) 2467.
- [4] S.B. Kim, J.S. Cho, *IEEE Photon. Technol. Lett.* 9 (2) (1997) 161.
- [5] J.I. Kinoshita, K. Matsumoto, *IEEE J. Quantum Electron.* 24 (11) (1988) 2160.
- [6] J.E.A. Whiteaway, G.H.B. Thompson, A.J. Collar, C.J. Armistead, *IEEE J. Quantum Electron.* 25 (6) (1989) 1261.
- [7] J.I. Kinoshita, K. Matsumoto, *IEEE J. Quantum Electron.* 25 (6) (1989) 1324.
- [8] J.E.A. Whiteaway, B. Garret, G.B.H. Thompson, A.J. Collar, C.J. Armistead, M.J. Fice, *IEEE J. Quantum Electron.* 28 (5) (1992) 1277.
- [9] Y. Chen, R. Maciejko, T. Makino, *IEEE J. Quantum Electron.* 34 (1) (1998) 113.
- [10] F. Randone, I. Montrosset, *IEEE J. Quantum Electron.* 31 (11) (1995) 1964.
- [11] N. Sasa, *IEEE J. Quantum Electron.* 33 (12) (1997) 2255.
- [12] T. Fessant, *IEEE Proc. Optoelectron.* 144 (6) (1997) 365.
- [13] S.F. Yu, N.Q. Ngo, *IEEE J. Quantum Electron.* 38 (8) (2002) 1062.
- [14] G.P. Agrawal, N.K. Dutta, *Semiconductor Lasers*, Van Nostrand Reinhold, New York, 1993.
- [15] G. Toptchiyski, S. Kindt, K. Petermann, E. Hilliger, S. Diez, H. Weber, *IEEE/OSA J. Lightwave Technol.* 17 (12) (1999) 2577.
- [16] Y. Kim, J. Jeong, *IEEE J. Quantum Electron.* 39 (10) (2003) 1314.
- [17] Y. Kim, H. Lee, J. Lee, J. Han, T.W. Oh, J. Jeong, *IEEE J. Quantum Electron.* 36 (8) (2000) 900.
- [18] Z.M. Wu, G.Q. Xia, X.H. Jia, *IEEE J. Quantum Electron.* 41 (3) (2005) 384.
- [19] G.P. Agrawal, A.H. Bobeck, *IEEE J. Quantum Electron.* 24 (12) (1988) 2407.